CONSENTED VOICE SYNTHESIS & SPEECH-DECODER CALIBRATION SUBSET

Companion subset expansion — hypothetical research integration

Consent-based research & simulation

Concept architecture / Proposed links

External supply / battery

Controller power

Separate modality

Research EXG

Research host power

NeuroSkill

Acquisition → Features → Embeddings → Annotations

Model estimates

OPTIONAL CLOUD

API gateway → Authenticated transport → LLM endpoint → Optional TTS API → Model storage → Job queue

Ethernet / Wi-Fi

Cellular backhaul

Optional

PROPOSED IMPLANT INTERFACES

Acoustic sensing

Middle-ear motion

UmboMic → Charge amplifier → Acoustic digitizer → Device telemetry

Neural sensing

Experimental

Recording array → Neural amplifier → Neural digitizer → Device telemetry

DEVICE-SPECIFIC OUTPUT

Perceptual validation

L / R stimulators

L receiver / stimulator

R receiver / stimulator

L cochlear electrodes

R cochlear electrodes

Device power / charging

To be specified

EXTERNAL CONTROLLER

Deployment alternatives

Controller power bus

ACQUISITION & INTERPRETATION

ACOUSTIC

Device receiver → Time alignment → Signal quality → Audio features → Speech activity → Speech recognition → Recognized speech

Proposed adapter

NEURAL RESEARCH

Participant-specific

Experimental

Device receiver → Time alignment → Signal quality → Artifact rejection → Neural features → Task-trained decoder → Confidence gate

Approved model

Reviewed settings

EXG context

Model estimates

Reject / no output

Candidate neural output

VOLITIONAL SPEECH CALIBRATION

Participant-approved cueing

No impersonation

APPROVED CUE LIBRARY

Neutral prompt

Preference statement

Correction prompt

Recall cue

Scenario transition

Affective prosody

Controlled variant

Category tiles only

Approved variants — approx. 1500

Specified design inventory

Participant review

Consent gate

Real-time research loop

Context buffer → Prompt scheduler → Synthesis request → Latency monitor

Runtime context

Timing feedback

No fabricated values

MEMORY & RECORDS

Recorded events / Retrieval

Session history

Retrieval index

Scenario state

Voice registry

Model versions

Provenance

Retention

Audit log

OUTPUT & PERCEPTION

Ear canal

Speaker / headphones

Acoustic playback

Auditory perception

Interface validation

Stimulation encoder

Device speech processor

CI audio interface

L

R

Compatible telemetry

Physical routes / Perceptual assessment

CONSENTED VOICE LOOP

Offline preparation

Participant text approval

Approved script text

Application text

Authorized recordings

Audio quality

Voice enrollment

Voice-owner consent

Voice profile

Model reference

Live microphone

Voice profile

Speaker-conditioned TTS

Real-time voice conversion

Optional / consented

Approved clips

Clip playback

Affective prosody — participant-approved

Disclosure banner

Participant-visible display

Runtime rendering

Prosody

Pacing

Pronunciation

Voice ID

Optional mix inputs

Selected music

Level

Ambient audio

Level

Playback timing

Source selector

Audio chunks

Playback buffer

Turn arbiter

Level limiter

Audio mix

Mute

Output monitor

Reviewed settings

Slower feedback

CALIBRATION & VALIDATION

Device status → Link quality → Health monitor

Physical measurements

Measurement interface → Recorded response → Response analysis → Calibration review

Response sensing

Timing markers

User feedback → Perceptual assessment → Approved settings

Reported percepts

Output interrupt

Participant control

Start / stop

Session limits

Fault mute

OFFLINE DEVELOPMENT

Labeled trials → Training → Held-out evaluation → Model registry → Approved update ← Rollback

Training datasets

Latency

Signal quality

Decoder error

Playback status

FUNCTION / DESIGN STATUS

Data

Audio

Control

Power

Proposed

Slow feedback

Schematic • Not to scale